

Fast gyrosynchrotron codes for arbitrary charged particles

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ABSTRACT

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1. EXACT FORMULAE

In a stationary medium, without scattering, the radiation transfer equation has the form

$$\frac{dI_f^\sigma(z)}{dz} = j_f^\sigma(z) - \kappa^\sigma(z)I_f^\sigma(z), \quad (1)$$

where I_f^σ is the intensity [erg cm⁻² s⁻¹ Hz⁻¹ sr⁻¹] of the wave mode σ , and z is the coordinate along the propagation direction.

Considering the emission and absorption of the magnetohydrodynamic waves due to interaction with charged particles in a magnetic field, and following D. B. Melrose (1968, 1980), we can write the exact expressions for the emissivity j_f^σ and absorption coefficient κ^σ as

$$j_f^\sigma = \frac{2\pi q^2}{c} \frac{n_\sigma f^2}{1 + T_\sigma^2} \times \sum_{s=-\infty}^{\infty} \int \left[\frac{T_\sigma(\cos\theta - n_\sigma\beta\mu) + L_\sigma \sin\theta}{n_\sigma \sin\theta} J_s(\lambda) - \epsilon\beta\sqrt{1-\mu^2} J_s'(\lambda) \right]^2 F(\mathbf{p}) \times \delta\left(f - f n_\sigma\beta\mu \cos\theta - \frac{sf_B}{\gamma}\right) d^3\mathbf{p}, \quad (2)$$

$$\kappa^\sigma = -\frac{2\pi q^2}{n_\sigma(1 + T_\sigma^2)} \times \sum_{s=-\infty}^{\infty} \int \left[\frac{T_\sigma(\cos\theta - n_\sigma\beta\mu) + L_\sigma \sin\theta}{n_\sigma \sin\theta} J_s(\lambda) - \epsilon\beta\sqrt{1-\mu^2} J_s'(\lambda) \right]^2 \times \frac{1}{\beta} \left[\frac{\partial F(\mathbf{p})}{\partial p} + \frac{n_\sigma\beta \cos\theta - \mu}{p} \frac{\partial F(\mathbf{p})}{\partial \mu} \right] \times \delta\left(f - f n_\sigma\beta\mu \cos\theta - \frac{sf_B}{\gamma}\right) d^3\mathbf{p}, \quad (3)$$

where f is the emission frequency, f_B is the cyclotron frequency,

$$f_B = \frac{|q|B}{2\pi mc} \quad (4)$$

(note that the cyclotron frequency is always positive), q and m are respectively the charge and mass of the considered particles, B is the magnetic field strength, c is the speed of light, n_σ , T_σ , and L_σ are respectively the refraction index and two components of the polarization vector of the considered waves (see Appendix A), θ is the propagation direction (the angle between the wave vector and the magnetic field vector), p is the particle momentum, $\beta = p/(\gamma mc)$ is the normalized (by c) particle speed, γ is the relativistic factor,

$$\gamma = \sqrt{1 + \frac{p^2}{m^2 c^2}} = \frac{1}{\sqrt{1 - \beta^2}}, \quad (5)$$

$\mu = \cos\alpha$, α is the pitch angle (the angle between the particle momentum and the magnetic field vector), $F(\mathbf{p})$ is the distribution function of the considered particles, $J_s(\lambda)$ and $J_s'(\lambda)$ are respectively the Bessel function and its derivative over the argument λ ,

$$\lambda = \frac{f}{f_B} \gamma n_\sigma \beta \sin\theta \sqrt{1 - \mu^2}, \quad (6)$$

and the parameter $\epsilon = |q|/q$ specifies the gyration direction of the considered particles: $\epsilon = -1$ for electrons and muons; $\epsilon = +1$ for positrons, protons, and α -particles. The distribution function $F(\mathbf{p})$ satisfies the normalization condition

$$\int F(\mathbf{p}) d^3\mathbf{p} = N, \quad (7)$$

where N is the number density of the considered particles.

In astrophysical applications, it is often convenient to express the distribution function in terms of the particle energy E and pitch angle α (or μ). The distribution function $F(E, \mu)$, which is assumed to be azimuthally symmetric, satisfies the normalization condition

$$2\pi \int_{E_{\min}}^{E_{\max}} dE \int_{-1}^1 F(E, \mu) d\mu = N, \quad (8)$$

where $E_{\min} \leq E \leq E_{\max}$ is the energy range of the considered particles. The corresponding expressions for the emissivity j_f^σ and absorption coefficient κ^σ take the form

$$j_f^\sigma = \frac{4\pi^2 q^2}{c} \frac{n_\sigma f^2}{1 + T_\sigma^2} \sum_{s=-\infty}^{\infty} \int_{E_{\min}}^{E_{\max}} dE \times \int_{-1}^1 \left[\frac{T_\sigma(\cos\theta - n_\sigma \beta \mu) + L_\sigma \sin\theta}{n_\sigma \sin\theta} J_s(\lambda) - \epsilon \beta \sqrt{1 - \mu^2} J'_s(\lambda) \right]^2 F(E, \mu) \times \delta \left(f - f n_\sigma \beta \mu \cos\theta - \frac{s f_B}{\gamma} \right) d\mu, \quad (9)$$

$$\kappa^\sigma = -\frac{4\pi^2 q^2 c}{n_\sigma (1 + T_\sigma^2)} \sum_{s=-\infty}^{\infty} \int_{E_{\min}}^{E_{\max}} dE \times \int_{-1}^1 \left[\frac{T_\sigma(\cos\theta - n_\sigma \beta \mu) + L_\sigma \sin\theta}{n_\sigma \sin\theta} J_s(\lambda) - \epsilon \beta \sqrt{1 - \mu^2} J'_s(\lambda) \right]^2 \left[\frac{\partial F(E, \mu)}{\partial E} - \frac{1 + \beta^2}{c p \beta} F(E, \mu) + \frac{n_\sigma \beta \cos\theta - \mu}{c p \beta} \frac{\partial F(E, \mu)}{\partial \mu} \right] \times \delta \left(f - f n_\sigma \beta \mu \cos\theta - \frac{s f_B}{\gamma} \right) d\mu. \quad (10)$$

2. APPROXIMATE FORMULAE

Following the approach by [G. D. Fleishman & A. A. Kuznetsov \(2010\)](#), we replace summation over the discrete harmonic number s in the above expressions by

integration over a corresponding continuous parameter, i.e.,

$$\sum_{s=-\infty}^{\infty} U(s) \simeq \int_{-\infty}^{\infty} U(s) ds. \quad (11)$$

Now, the expressions for j_f^σ and κ^σ become triple integrals over dE , $d\mu$, and ds . The integral over ds can be computed using the δ -function. As the result, from (9–10) we obtain

$$j_f \simeq \frac{4\pi^2 q^2}{c f_B} \frac{n f^2}{1 + T^2} \int_{E_{\min}}^{E_{\max}} \gamma dE \times \int_{-1}^1 \left[\frac{T(\cos\theta - n\beta\mu) + L \sin\theta}{n \sin\theta} J_s(\lambda) - \epsilon \beta \sqrt{1 - \mu^2} J'_s(\lambda) \right]^2 F(E, \mu) d\mu, \quad (12)$$

$$\kappa \simeq -\frac{4\pi^2 q^2 c}{n f_B (1 + T^2)} \int_{E_{\min}}^{E_{\max}} \gamma dE \times \int_{-1}^1 \left[\frac{T(\cos\theta - n\beta\mu) + L \sin\theta}{n \sin\theta} J_s(\lambda) - \epsilon \beta \sqrt{1 - \mu^2} J'_s(\lambda) \right]^2 \left[\frac{\partial F(E, \mu)}{\partial E} - \frac{1 + \beta^2}{c p \beta} F(E, \mu) + \frac{n \beta \cos\theta - \mu}{c p \beta} \frac{\partial F(E, \mu)}{\partial \mu} \right] d\mu, \quad (13)$$

where we omitted the wave-mode indices σ for brevity, and the index s of the Bessel functions is determined from the resonance condition:

$$s = s(E, \mu) = \frac{f}{f_B} \gamma (1 - n \beta \mu \cos\theta). \quad (14)$$

For the Bessel functions, we use the approximate expressions ([J. P. Wild & E. R. Hill 1971](#)):

$$J_s(sx) \simeq \frac{1}{\sqrt{2\pi s}} \frac{Z^s(x)}{a(s, x)}, \quad J'_s(sx) \simeq \frac{b(s, x)}{\sqrt{2\pi s}} \frac{Z^s(x)}{x}, \quad (15)$$

where $sx = \lambda$ and

$$Z(x) = \frac{x \exp(\sqrt{1 - x^2})}{1 + \sqrt{1 - x^2}}, \quad (16)$$

$$a(s, x) = \left[(1 - x^2)^{3/2} + \frac{A}{s} \right]^{1/6}, \quad A = 0.503297, \quad (17)$$

$$b(s, x) = \left[(1 - x^2)^{3/2} + \frac{B}{s} \right]^{1/6} \left(1 - \frac{1}{5s^{2/3}} \right),$$

$$B = 1.193000. \quad (18)$$

Since s is determined by the Equation (14), the parameter x takes the form

$$x(E, \mu) = \frac{\lambda}{s} = \frac{n\beta \sin \theta \sqrt{1 - \mu^2}}{1 - n\beta \mu \cos \theta}. \quad (19)$$

Substituting the above approximate expressions into (12–13), we obtain

$$j_f \simeq \frac{2\pi q^2}{c} \frac{f}{n(1 + T^2) \sin^2 \theta} \int_{E_{\min}}^{E_{\max}} dE$$

$$\times \int_{-1}^1 [Z(E, \mu)]^{2s(E, \mu)} Q(E, \mu) F(E, \mu) d\mu, \quad (20)$$

$$\kappa \simeq -\frac{2\pi q^2 c}{n^3 f (1 + T^2) \sin^2 \theta} \int_{E_{\min}}^{E_{\max}} dE$$

$$\times \int_{-1}^1 [Z(E, \mu)]^{2s(E, \mu)} Q(E, \mu) R(E, \mu) d\mu, \quad (21)$$

where

$$Q(E, \mu) = [T(\cos \theta - n\beta \mu) + L \sin \theta$$

$$- \epsilon ab(1 - n\beta \mu \cos \theta)]^2$$

$$\times [a^2(1 - n\beta \mu \cos \theta)]^{-1}, \quad (22)$$

$$R(E, \mu) = \frac{\partial F(E, \mu)}{\partial E}$$

$$- \frac{1 + \beta^2}{cp\beta} F(E, \mu) + \frac{n\beta \cos \theta - \mu}{cp\beta} \frac{\partial F(E, \mu)}{\partial \mu}. \quad (23)$$

Figure 1 presents examples of the integrand over $d\mu$ in Equation (20), i.e., plots of $Z^{2s} Q F$ vs. μ for a fixed energy. The following parameters are used: magnetic field strength $B = 270$ G, electron number density (that determines the wave dispersion, see Appendix A) $N_{e-} = 5 \times 10^9 \text{ cm}^{-3}$, emission frequency $f = 10$ GHz, viewing angle $\theta = 75^\circ$, energy of the non-thermal particles $E = 1$ MeV; electrons and positrons are considered. The distribution function is assumed to be isotropic and equal unity. It follows from the figure that electrons produce slightly more X-mode than O-mode emission, while positrons favour O-mode.

Figure 2 presents similar examples of the integrand over $d\mu$ in Equation (20) for protons and antiprotons. The following parameters are used: magnetic field strength $B = 1500$ G, electron number density $N_{e-} = 10^8 \text{ cm}^{-3}$, emission frequency $f = 10$ GHz, viewing angle $\theta = 80^\circ$, energy of the non-thermal particles $E = 10$ GeV, the distribution function is assumed to be isotropic and equal unity. Now, the contributions of protons and antiprotons prevail respectively for O-mode and X-mode; however, for both types of particles the X-mode emission is suppressed (due to the Razin effect) in comparison with the O-mode emission, so that O-mode should always dominate. Note also that the integrand values are much smaller than those for electrons and positrons.

Since the integrands over $d\mu$ in the expressions (20–21) have sharp peaks (see Figures 1–2), we can estimate the corresponding integrals using the Laplace method, which results in the following approximate equations:

$$j_f \simeq \frac{2\pi q^2}{c} \frac{f}{n(1 + T^2) \sin^2 \theta} \int_{E_{\min}}^{E_{\max}} [Z(E, \mu_0)]^{2s(E, \mu_0)}$$

$$\times Q(E, \mu_0) f(E, \mu_0) \sqrt{-\frac{2\pi}{h''_{\mu\mu}(E, \mu_0)}} dE, \quad (24)$$

$$\kappa \simeq -\frac{2\pi q^2 c}{n^3 f (1 + T^2) \sin^2 \theta} \int_{E_{\min}}^{E_{\max}} [Z(E, \mu_0)]^{2s(E, \mu_0)}$$

$$\times Q(E, \mu_0) R(E, \mu_0) \sqrt{-\frac{2\pi}{h''_{\mu\mu}(E, \mu_0)}} dE, \quad (25)$$

where h is the logarithm of the integrand over $d\mu$ in Equation (20):

$$h(E, \mu) = 2s(E, \mu) \ln Z(E, \mu)$$

$$+ \ln Q(E, \mu) + \ln f(E, \mu), \quad (26)$$

and all quantities in (24–25) are computed at the value of $\mu_0 = \mu_0(E)$ where the function $h(E, \mu)$ reaches its maximum as a function of μ for a given energy E .

Finding the maximum of $h(E, \mu)$ is reduced to solving the equation $h'_\mu(E, \mu) = 0$, where $h'_\mu(E, \mu)$ is the derivative of the function $h(E, \mu)$ over μ :

$$h'_\mu(E, \mu) = 2\gamma \frac{f}{f_B} \left(\frac{n\beta \cos \theta - \mu}{1 - \mu^2} \sqrt{1 - x^2} \right.$$

$$\left. - n\beta \cos \theta \ln Z \right) + [\ln Q(E, \mu)]'_\mu + \frac{f'_\mu(E, \mu)}{f(E, \mu)}, \quad (27)$$

where

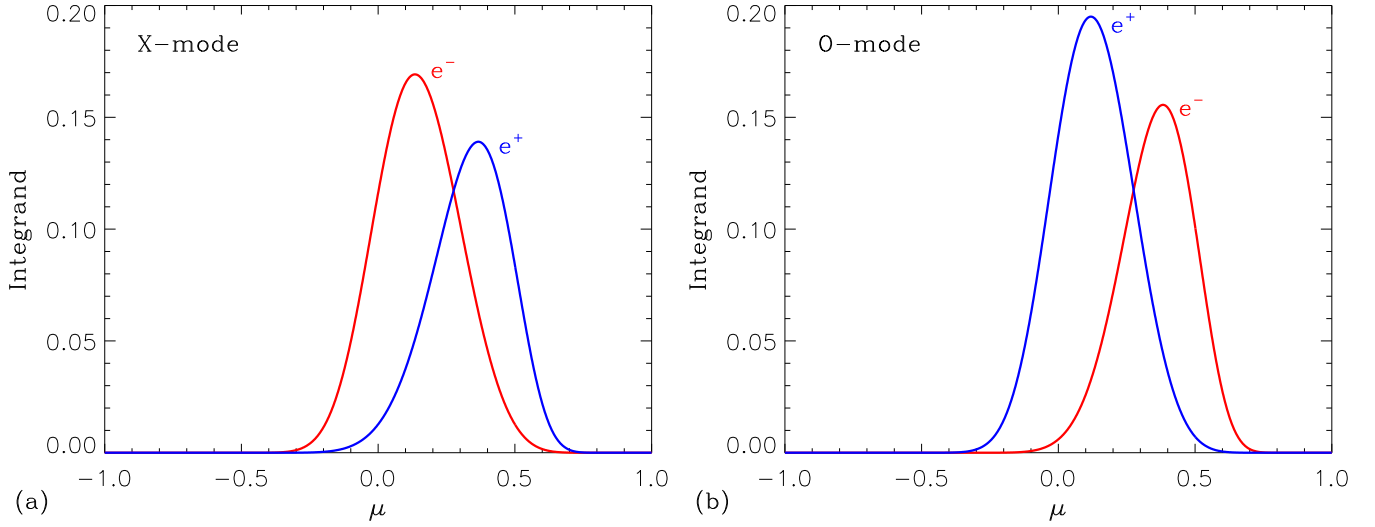


Figure 1. Integrand over the pitch-angle cosine in the expression (20) for the gyrosynchrotron emissivity, for electrons and positrons (see text for the model parameters).

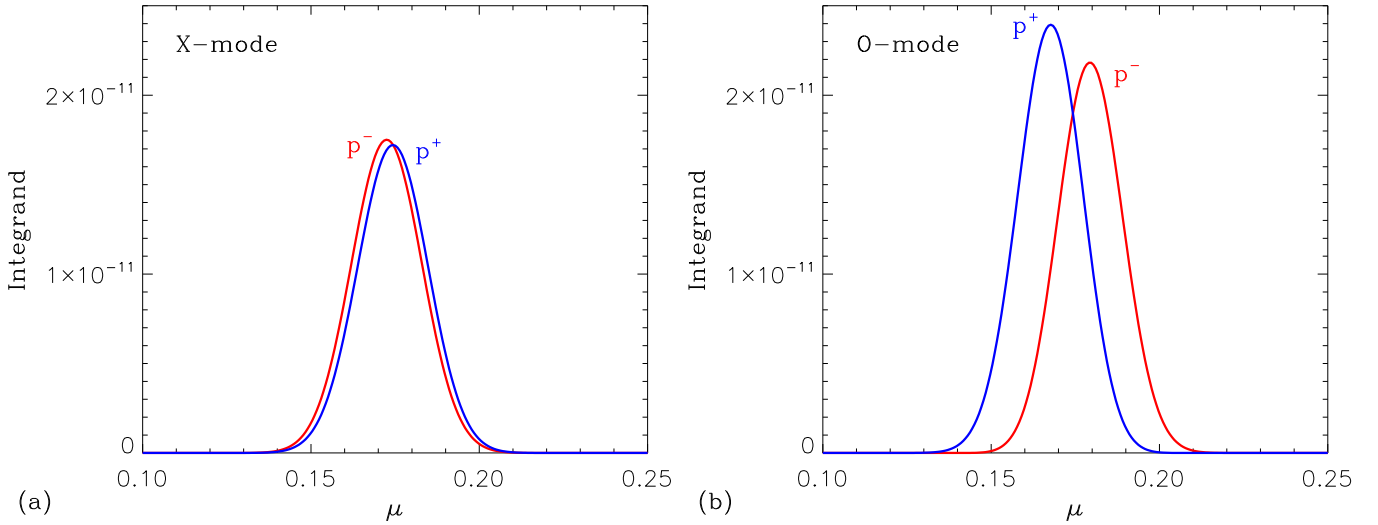


Figure 2. Integrand over the pitch-angle cosine in the expression (20) for the gyrosynchrotron emissivity, for protons and antiprotons (see text for the model parameters).

$$\begin{aligned}
 [\ln Q(E, \mu)]'_\mu &= 2 \left\{ \epsilon ab \left[n\beta \cos \theta \right. \right. \\
 &\quad \left. \left. - \left(\frac{a'_\mu}{a} + \frac{b'_\mu}{b} \right) (1 - n\beta \mu \cos \theta) \right] - T n\beta \right\} \\
 &\times [T(\cos \theta - n\beta \mu) + L \sin \theta - \epsilon ab(1 - n\beta \mu \cos \theta)]^{-1} \\
 &\quad - 2 \frac{a'_\mu}{a} + \frac{n\beta \cos \theta}{1 - n\beta \mu \cos \theta}. \quad (28)
 \end{aligned}$$

The derivatives of the parameters a and b in the above expression can be computed as

$$\frac{a'_\mu}{a} = \frac{\chi + A\eta}{6a_1}, \quad (29)$$

$$\frac{b'_\mu}{b} = \frac{\chi + B\eta}{6b_1} - \frac{2}{15} \frac{\eta s^{1/3}}{b_2}, \quad (30)$$

$$a_1 = (1 - x^2)^{3/2} + \frac{A}{s}, \quad a = a_1^{1/6}, \quad (31)$$

$$\begin{aligned}
 b_1 &= (1 - x^2)^{3/2} + \frac{B}{s}, \quad b_2 = 1 - \frac{1}{5s^{2/3}}, \\
 b &= b_1^{1/6} b_2, \quad (32)
 \end{aligned}$$

$$\chi = 3x\sqrt{1 - x^2} \frac{\mu - n\beta \cos \theta}{(1 - n\beta \mu \cos \theta)^2} \frac{n\beta \sin \theta}{\sqrt{1 - \mu^2}}, \quad (33)$$

$$\eta = \frac{f}{f_B} \frac{\gamma n\beta \cos \theta}{s^2}. \quad (34)$$

The second derivative of $h(E, \mu)$ over μ has the form

$$\begin{aligned}
h''_{\mu\mu}(E, \mu) = & -2\Gamma \frac{f}{f_B} \\
& \times \frac{1 + \mu^2 - 4N\beta\mu \cos \theta + N^2\beta^2(\mu^2 + \cos 2\theta)}{(1 - \mu^2)^2(1 - N\beta\mu \cos \theta)\sqrt{1 - x^2}} + \\
& + [\ln Q(E, \mu)]''_{\mu\mu} + \frac{f''_{\mu\mu}(E, \mu)}{f(E, \mu)} - \left[\frac{f'_\mu(E, \mu)}{f(E, \mu)} \right]^2, \quad (35)
\end{aligned}$$

where it is more convenient to calculate $[\ln Q(E, \mu)]''_{\mu\mu}$ numerically.

As expected, for $\epsilon = -1$ (i.e., for electrons) all the above formulae become equivalent to those in the paper of [G. D. Fleishman & A. A. Kuznetsov \(2010\)](#).

3. IMPLEMENTATION

We adopt a factorized form of the particle distribution function:

$$F(E, \mu) = F_1(E)F_2(\mu), \quad (36)$$

where the distributions over energy and pitch-angle cosine satisfy the normalization conditions

$$2\pi \int_{E_{\min}}^{E_{\max}} F_1(E) dE = N, \quad \int_{-1}^1 F_2(\mu) d\mu = 1. \quad (37)$$

Plasma can contain several populations of emitting particles. In such case, the emissivity and absorption coefficient are computed as

$$j_f^\sigma = \sum_i (j_f^\sigma)_i, \quad \kappa^\sigma = \sum_i (\kappa^\sigma)_i, \quad (38)$$

where $(j_f^\sigma)_i$ and $(\kappa^\sigma)_i$ are respectively the emissivity and absorption coefficient due to the i -th plasma component. Optionally, the free-free emissivity and absorption coefficient $(j_f^\sigma)_{\text{FF}}$ and $(\kappa^\sigma)_{\text{FF}}$ (determined by the thermal electron and ion plasma components only, and computed using the approximate formulae from [G. A. Dulk 1985](#)) can be added, too.

We consider the emission produced in a homogeneous source and observed from the Sun-Earth distance. The corresponding solution of the radiation transfer equation (1), corrected for the distance, has the form

$$I_f^\sigma = \frac{S}{\mathcal{R}^2} \frac{j_f^\sigma}{\kappa^\sigma} \left(1 - e^{-\kappa^\sigma \Delta z} \right), \quad (39)$$

where S is the visible source area, Δz is the source depth along the line-of-sight, and $\mathcal{R}$ is the astronomical unit.

4. RESULTS

Figure 3 demonstrates examples of the gyrosynchrotron emission spectra for electrons and positrons. In both cases, the source parameters are: source area $S = (20'')^2$, source depth $\Delta z = 15''$, thermal electron concentration $N_{e-} = 5 \times 10^9 \text{ cm}^{-3}$, magnetic field $B = 270 \text{ G}$, viewing angle $\theta = 75^\circ$, the free-free emission is ignored. The energetic particle parameters are: concentration $N_{e-} = N_{e+} = 10^6 \text{ cm}^{-3}$, single power-law energy distribution in the range from $E_{\min} = 100 \text{ keV}$ to $E_{\max} = 10 \text{ MeV}$ with the index of $\delta = 4$, isotropic pitch-angle distribution. As expected, the optically thin emission is polarized in X-mode for electrons and in O-mode for positrons (for electrons, the polarization degree is slightly higher).

Figure 4 demonstrates examples of the gyrosynchrotron emission spectra for protons and antiprotons. In both cases, the source parameters are: source area $S = (20'')^2$, source depth $\Delta z = 15''$, thermal electron concentration $N_{e-} = 10^8 \text{ cm}^{-3}$, magnetic field $B = 1500 \text{ G}$, viewing angle $\theta = 80^\circ$, the free-free emission is ignored. The energetic particle parameters are: concentration $N_{p+} = N_{p-} = 3 \times 10^7 \text{ cm}^{-3}$, single power-law energy distribution in the range from $E_{\min} = 2 \text{ MeV}$ to $E_{\max} = 70 \text{ GeV}$ with the index of $\delta = 1.5$, isotropic pitch-angle distribution. The resulting spectra for the protons and antiprotons are almost the same (for antiprotons, the X-mode emission is slightly stronger and the O-mode emission is slightly weaker than for protons).

APPENDIX

A. DISPERSION OF THE MAGNETOIONIC MODES

We assume that the dispersion characteristics of the magnetoionic waves are determined by the electron component only, because concentrations of other components are usually negligible. The refraction index of the electromagnetic waves in a magnetized plasma satisfies

the dispersion equation

$$n_\sigma^2 = 1 - \frac{2v(1-v)}{2(1-v) - u \sin^2 \theta + \sigma \sqrt{D}}, \quad (A1)$$

where

$$D = u^2 \sin^4 \theta + 4u(1-v)^2 \cos^2 \theta, \quad (A2)$$

$$u = (f_{\text{Be}}/f)^2, \quad v = (f_{\text{pe}}/f)^2, \quad (A3)$$

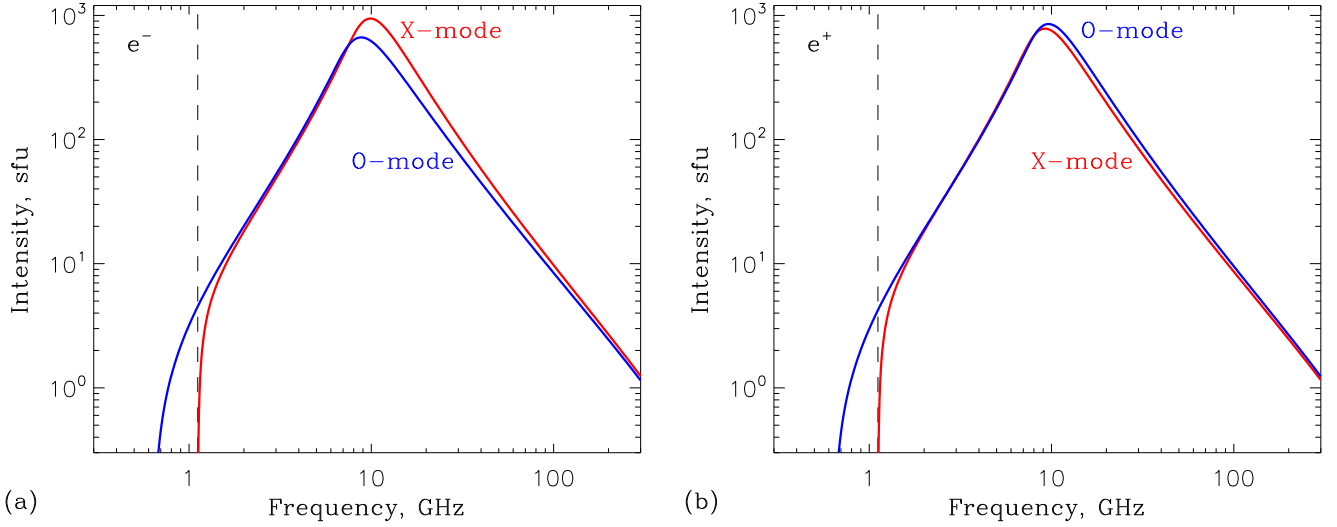


Figure 3. Spectra of gyrosynchrotron emission from relativistic electrons (a) and positrons (b). The vertical dashed line marks the X-mode cutoff frequency. See text for the model parameters.

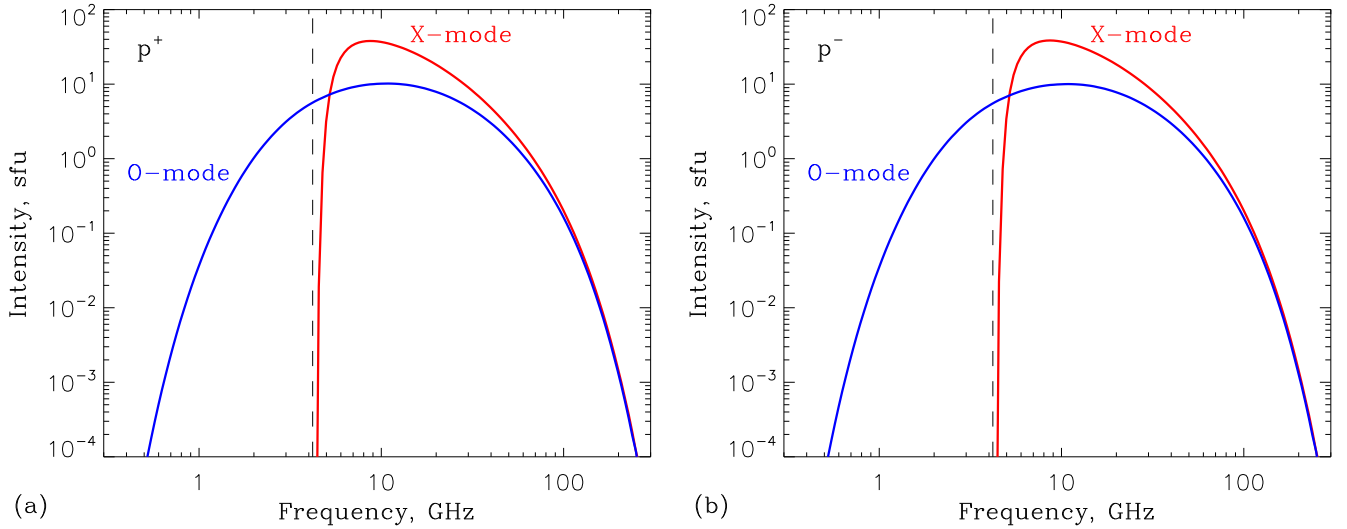


Figure 4. Spectra of gyrosynchrotron emission from ultrarelativistic protons (a) and antiprotons (b). The vertical dashed line marks the X-mode cutoff frequency. See text for the model parameters.

f_{Be} and f_{pe} are respectively the electron cyclotron and electron plasma frequencies,

$$f_{Be} = \frac{|q_e|B}{2\pi m_e c}, \quad f_{pe} = \sqrt{\frac{q_e^2 N_{e^-}}{\pi m_e}}, \quad (A4)$$

and N_{e^-} is the total electron number density. The parameter σ determines the wave mode: $\sigma = -1$ for X-mode; $\sigma = +1$ for O-mode.

In the reference frame where the z axis is directed along the magnetic field and the wave vector $\mathbf{k}$ lies in the xz plane, the polarization vector of the waves has the form

$$\mathbf{e}_\sigma = (T_\sigma \cos \theta + L_\sigma \sin \theta, i, -T_\sigma \sin \theta + L_\sigma \cos \theta) \times (1 + T_\sigma^2 + L_\sigma^2)^{-1/2}, \quad (A5)$$

where

$$T_\sigma = \frac{2\sqrt{u}(1-v)\cos\theta}{u\sin^2\theta - \sigma\sqrt{D}}, \quad (A6)$$

$$L_\sigma = \frac{v\sqrt{u}\sin\theta + T_\sigma uv\sin\theta\cos\theta}{1 - u - v + uv\cos^2\theta}. \quad (A7)$$

Electromagnetic waves can propagate in plasma if their frequency exceeds the cutoff frequency, $f > f_c^\sigma$, where

$$f_c^O = f_{pe}, \quad f_c^X = \frac{f_{Be}}{2} + \sqrt{f_{pe}^2 + \frac{f_{Be}^2}{4}}. \quad (A8)$$

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